# AQ7.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.
**Level 1**

*1 point*
Choose the set of correct options.

- [ ] If a $3 \times 33 \times 3$ matrix $A$A is similar to the identity matrix of order 33, then $A$A must be the identity matrix of order 3.
- [ ] If a $3 \times 33 \times 3$ matrix $A$A is similar to a diagonal matrix of order 33, then $A$A must be a diagonal matrix of order 3.
- [ ] If $A$A and $B$B are similar matrices, then they are also equivalent matrices.
- [ ] If $A$A and $B$B are equivalent matrices, then they are also similar matrices.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
If a $3 \times 33 \times 3$ matrix $A$A is similar to the identity matrix of order 33, then $A$A must be the identity matrix of order 3.
If $A$A and $B$B are similar matrices, then they are also equivalent matrices.

Consider the linear transformation given below:
$T: R^3 \rightarrow R^2$ T : R3 → R2

$T(x, y, z) = (x - y, y + z)$T(x, y, z) = (x − y, y + z)

Consider two ordered bases $\beta_1 = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$β1 = {(1, 0, 0), (0, 1, 0), (0, 0, 1)}, and $\beta_2 = \{(1, 0, 1), (0, 1, 1), (1, 1, 0)\}$β2 = {(1, 0, 1), (0, 1, 1), (1, 1, 0)} of $R^3$ R3 and consider the standard ordered basis $\gamma = \{(1, 0), (0, 1)\}$γ = {(1, 0), (0, 1)} of $R^2$ R2. Let $A$A and $B$B be the matrices corresponding to the linear transformation $T$T with respect to the bases $\beta_1$ β1 and $\beta_2$ β2 for the domain, respectively and $\gamma$γ for the co-domain. Let $Q$Q and $P$P be matrices such that $B = QAP$B = QAP.

Answer the questions 2,3, and 4 using the above information.
*1 point*
Which of the following matrices represents $A$A?

- ○ $\begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 1 \end{bmatrix}$[11−1201]
- ○ $\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$[10−1101]
- ○ $\begin{bmatrix} 1 & 1 \\ -1 & 1 \\ 0 & 1 \end{bmatrix}$ $\begin{bmatrix} & & \\ & & \\ 1 & -1 & 0 & 1 & 1 & 1 \end{bmatrix}$
- ○

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ -1 & 2 & 1 \\ 1 & -1 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 & 1 & 0 & 1 \end{bmatrix}$

*1 point*
Which of the following matrices can be $P$?

- There does not exist any matrix $P$ which satisfies the property $B = QAP$.
- $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 2 & 1 & 1 & 0 \end{bmatrix}$
- $\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 & 1 & -2 & 1 & 2 & 1 & -2 & 1 & 2 & 1 & 2 & 1 & 2 & 1 & -2 & 1 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 \end{bmatrix}$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 \end{bmatrix}$

*1 point*
Which of the following statements is true for $Q$?

- [ ] $Q$ can be the identity matrix of order 3.
- [ ] $Q$ can be the identity matrix of order 2.
- [ ] Whatever the matrix $P$ we choose, $Q$ is always unique.
- [ ] $Q$ can be the matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 & 0 \end{bmatrix}$.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**

$Q$Q can be the identity matrix of order 2.
$Q$Q can be the matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$[0110].

*1 point*
Choose the set of correct options.
☐ If two square matrices of the same order have the same determinants, then they must be similar to each other.
☐ Similar matrices have the same rank.
☐
If $A$A and $B$B are similar matrices, and $Ax = b$Ax = b has a unique solution, then $Bx = b$Bx = b also has a unique solution.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
Similar matrices have the same rank.
If $A$A and $B$B are similar matrices, and $Ax = b$Ax = b has a unique solution, then $Bx = b$Bx = b also has a unique solution.

*1 point*
Choose the set of correct options.
☐
If det$A$A or det$B$B is nonzero, then $AB$AB and $BA$BA are similar.
☐
$A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$A = [2513] and $B = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$B = [2132] are similar.

☐
$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$A = [2013] and $B = \begin{bmatrix} 1 & 1 & 1 \\ 0 & & 4 \end{bmatrix}$B = [10114] are similar.

☐
$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$A = [2012] and $B = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$B = [2002] are not similar.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
If det$A$A or det$B$B is nonzero, then $AB$AB and $BA$BA are similar.
$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$A = [2012] and $B = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$B = [2002] are not similar.

*1 point*
Which of the following options are correct?
☐ Two invertible matrices of same order are equivalent.
☐
$A = \begin{bmatrix} 1 & 7 & 6 \\ 2 & 1 & -1 \\ 5 & 2 & -3 \end{bmatrix}$A = $\begin{bmatrix} & & \\ & & \\ 1 & 2 & 5 & 7 & 1 & 2 & 6 & -1 & -3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 & -2 \\ 1 & 1 & 4 \\ 1 & 2 & -6 \end{bmatrix}$B = $\begin{bmatrix} & & \\ & & \\ 2 & 1 & 1 & 3 & 1 & 2 & -2 & 4 & -6 \end{bmatrix}$ are equivalent.

☐
$A = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}$A = [2436] and $B = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$B = [1331] are equivalent.

☐

Consider two linear transformations $T_1(x, y, z) = (x + y - z, x + 4y - 2z, -2x + y + z)$T1 (x, y, z) = (x + y − z, x + 4y − 2z, −2x + y + z) and $T_2(x, y, z) = (x + y, x - z, y + z)$T2(x, y, z) = (x + y, x − z, y + z). Let $A$A and $B$B denote the matrix representation of $T_1$ T1 and $T_2$ T2 with respect to the standard basis of $R^3$ R3. Then $A$A and $B$B are equivalent.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Two invertible matrices of same order are equivalent.

$A = \begin{bmatrix} 17 & 6 \\ 21 & -1 \\ 5 & 2 & -3 \end{bmatrix}$A = $\begin{bmatrix} 1 & 2 & 5 & 7 & 1 & 2 & 6 & -1 & -3 \end{bmatrix}$ and $B = \begin{bmatrix} 23 & -2 \\ 11 & 4 \\ 12 & -6 \end{bmatrix}$B = $\begin{bmatrix} 2 & 11 & 3 & 12 & -24 & -6 \end{bmatrix}$ are equivalent.

Consider two linear transformations $T_1(x, y, z) = (x + y - z, x + 4y - 2z, -2x + y + z)$T1 (x, y, z) = (x + y − z, x + 4y − 2z, −2x + y + z) and $T_2(x, y, z) = (x + y, x - z, y + z)$T2(x, y, z) = (x + y, x − z, y + z). Let $A$A and $B$B denote the matrix representation of $T_1$ T1 and $T_2$ T2 with respect to the standard basis of $R^3$ R3. Then $A$A and $B$B are equivalent.

**Level 2**

*1 point*

Let $E$E be the set of $2 \times 2$2 × 2 matrices which are similar to a scalar matrix $A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$A = [3003].

Choose the set of correct statements.

- [ ] $E$E is an infinite set.
- [ ] Cardinality of $E$E is finite.
- [ ] $E$E is singleton set.
- [ ] $E$E consists of all the scalar matrices of order $n$n.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Cardinality of $E$E is finite.

$E$E is singleton set.

*1 point*

Which of the following option(s) is(are) true?

- [ ] If $A$ and $B$A and B are similar matrices then $A^{-1}$ and $B^{-1}$ A−1 and B−1 are similar matrices.
- [ ] If $A$ and $B$A and B are similar matrices then $A^2$ and $B^2$ A2 and B2 are similar matrices.
- [ ] If $A^2$ and $B^2$ A2 and B2 are similar matrices then $A$ and $B$A and B are similar matrices.
- [ ] If $A$ and $B$A and B are similar matrices then $A^T$ and $B^T$ AT and BT are similar matrices.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

If $A$ and $B$A and B are similar matrices then $A^{-1}$ and $B^{-1}$ A−1 and B−1 are similar matrices.

If $A$ and $B$A and B are similar matrices then $A^2$ and $B^2$ A2 and B2 are similar matrices.

If $A$ and $B$A and B are similar matrices then $A^T$ and $B^T$ AT and BT are similar matrices.
*1 point*
Let us consider the following matrices.

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \text{A} = [1101], \text{B} = [1001], \text{C} = [1011]$$

Choose the set of correct options.

- [ ] $A$A and $B$B have the same rank, and same determinant, but they are not similar matrices.
- [ ] $A$A and $C$C have the same rank, and same determinant, but they are not similar matrices.
- [ ] $B$B and $C$C have the same rank, and same determinant, but they are not similar matrices.
- [ ] None of the options are true.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$A$A and $B$B have the same rank, and same determinant, but they are not similar matrices.
$B$B and $C$C have the same rank, and same determinant, but they are not similar matrices.

Check Answers

Your score is: 0/10